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Course Welcome
Module 1 Introduction: What is Statistical Learning?
Lesson 1: Terminology and Types of Data
Lesson 2: Formal Description of Statistical Learning
Module 1 Summative Assessment
Graded Assignment: Module 1 Summative Assessment Submitted
Module 1 Summary

Your grade: 93.33%

Your latest: **93.33%** • Your highest: **93.33%** • To pass you need at least 80%. We keep your highest score.

Next item

Module 1 Summative Assessment

Review Learning Objectives

Instructions

1. What is the $EPE(f)$ **1 / 1 point**

- The average error in predictions.
- The exact error of each individual prediction.
- The difference between predicted and actual values.
- The total number of errors in your dataset.

Correct

2. What is a dataset? **1 / 1 point**

- A single data point
- A collection of related data from multiple samples.
- A statistical model
- An equation

Correct

3. What is the term for random fluctuations that cause the observed data to differ from the true model? **1 / 1 point**

- Bias
- Noise
- Variance
- Error

Correct

4. Which statistical learning model is used for predicting a continuous outcome variable? **1 / 1 point**

- Classification
- Regression
- Clustering
- Dimensionality Reduction

Correct

5. Which element of a statistical learning problem involves drawing conclusions based on the models? **1 / 1 point**

- Dataset
- Data
- Model
- Inference

Correct

6. Which data type is crucial for temporal order and may have autocorrelation? **1 / 1 point**

- Panel Data
- Numerical Data
- Cross-sectional Data
- Time-series Data

Correct

7. What is the purpose of prediction in statistical learning? **1 / 1 point**

- To make inferences, such as to estimate outcomes for new or unseen data.
- To create models
- To eliminate noise from data
- Performing t-tests on datasets

Correct

8. In which data type do images consist of pixels? **1 / 1 point**

- Image and Video Data
- Numerical Data
- Text Data
- Geospatial Data

Correct
Correct! Images in this data type consist of pixels and videos are sequences of images.

9. In which type of statistical learning problem is the focus on discovering patterns without labeled outcomes? **1 / 1 point**

- Regression
- Supervised Learning
- Classification
- Unsupervised Learning

Correct

10. Which of the following best describes the irreducible error in the context of EPE? **1 / 1 point**

- It can be reduced with a better model.
- B: It is the difference between the predicted and actual values.
- It arises from the model's inability to capture the relationship between X and Y.
- It arises from the inherent variability and noise in the data and cannot be eliminated, regardless of the model used.

Correct

11. Which of the following data types does not have an inherent order to its categories? **0 / 1 point**

- Ordinal Categorical Data
- Time-series Data
- Nominal Categorical Data
- Longitudinal Data

Incorrect
Ordinal categorical data has categories that are ordered.

12. What influences the kind of statistical learning models we can consider? **1 / 1 point**

- Information available on the dependent variable.
- Complexity of the dataset.
- The accuracy of the data.
- Number of independent variables.

Correct

13. In supervised learning, what kind of data is used to train models? **1 / 1 point**

- Unlabeled data
- Experimental data
- High-dimensional data
- Labeled data

Correct

14. Which of the following is true about time series data? **1 / 1 point**

- The presence of autocorrelation
- It is always continuous and numerical
- The lack of temporal order
- The number of data categories

Correct

15. What is the primary purpose of the clustering model? **1 / 1 point**

- Predicting a categorical outcome.
- Grouping similar data points together.
- Predicting a continuous outcome.
- Reducing the complexity of high-dimensional data.

Correct

coach
Ready to review what you've learned before starting the assignment? I'm here to help.
Help me practice Let's chat
Assignment details
Due Oct 27, 11:59 PM IST
Attempts 0 left (1 attempts every 8 hours)
Submitted Nov 14, 3:05 AM IST
Resume
Your grade
To pass you need at least 80%. We keep your highest score.
93.33%
View submission See feedback
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